Project _Insurance

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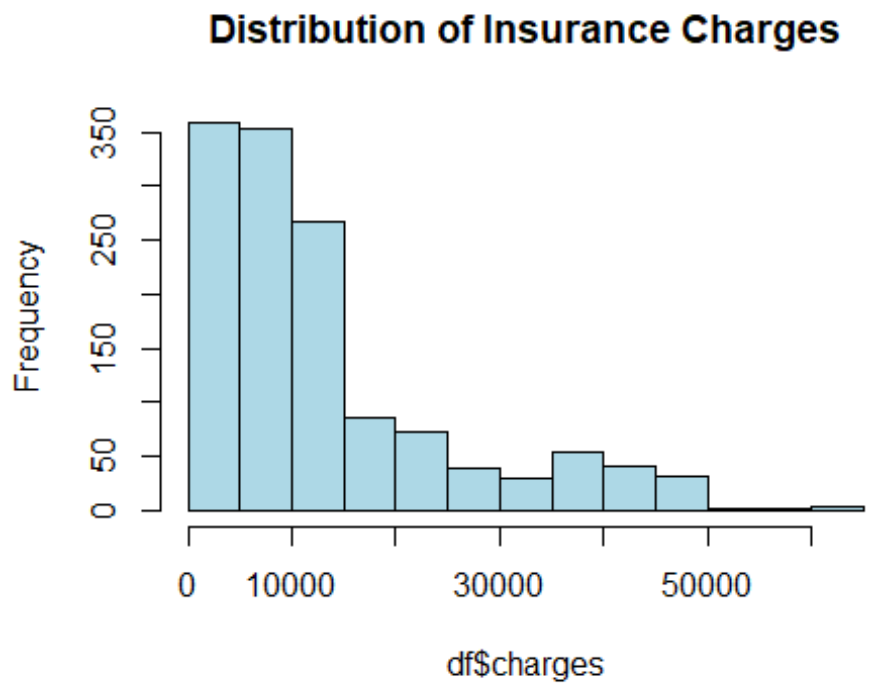
```r
df <- read.csv("C:/Users/HP/Downloads/insurance (1).csv")
str(df)
```

```
## 'data.frame':    1338 obs. of  7 variables:
##  $ age     : int  19 18 28 33 32 31 46 37 37 60 ...
##  $ sex     : chr  "female" "male" "male" "male" ...
##  $ bmi     : num  27.9 33.8 33 22.7 28.9 ...
##  $ children: int  0 1 3 0 0 0 1 3 2 0 ...
##  $ smoker  : chr  "yes" "no" "no" "no" ...
##  $ region  : chr  "southwest" "southeast" "southeast" "northwest" ...
##  $ charges : num  16885 1726 4449 21984 3867 ...
```

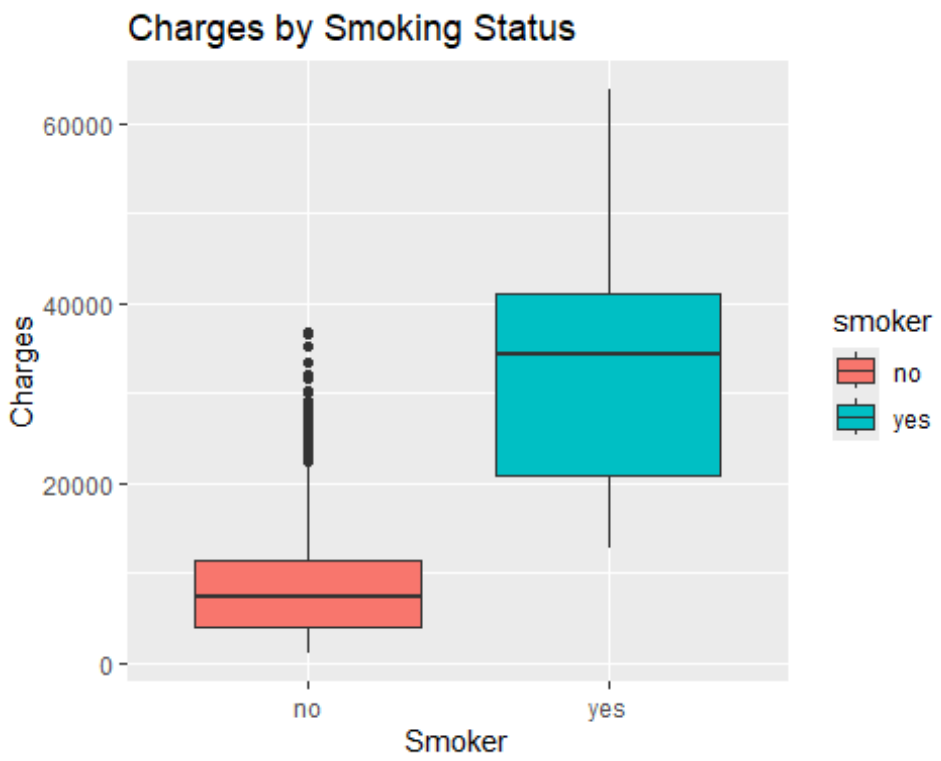
```r
summary(df)
```

```
##       age            sex                 bmi           children
##  Min.   :18.00   Length:1338        Min.   :15.96   Min.   :0.000
##  1st Qu.:27.00   Class :character   1st Qu.:26.30   1st Qu.:0.000
##  Median :39.00   Mode  :character   Median :30.40   Median :1.000
##  Mean   :39.21                      Mean   :30.66   Mean   :1.095
##  3rd Qu.:51.00                      3rd Qu.:34.69   3rd Qu.:2.000
##  Max.   :64.00                      Max.   :53.13   Max.   :5.000
##     smoker             region             charges
##  Length:1338        Length:1338        Min.   : 1122
##  Class :character   Class :character   1st Qu.: 4740
##  Mode  :character   Mode  :character   Median : 9382
##                                        Mean   :13270
##                                        3rd Qu.:16640
##                                        Max.   :63770
```

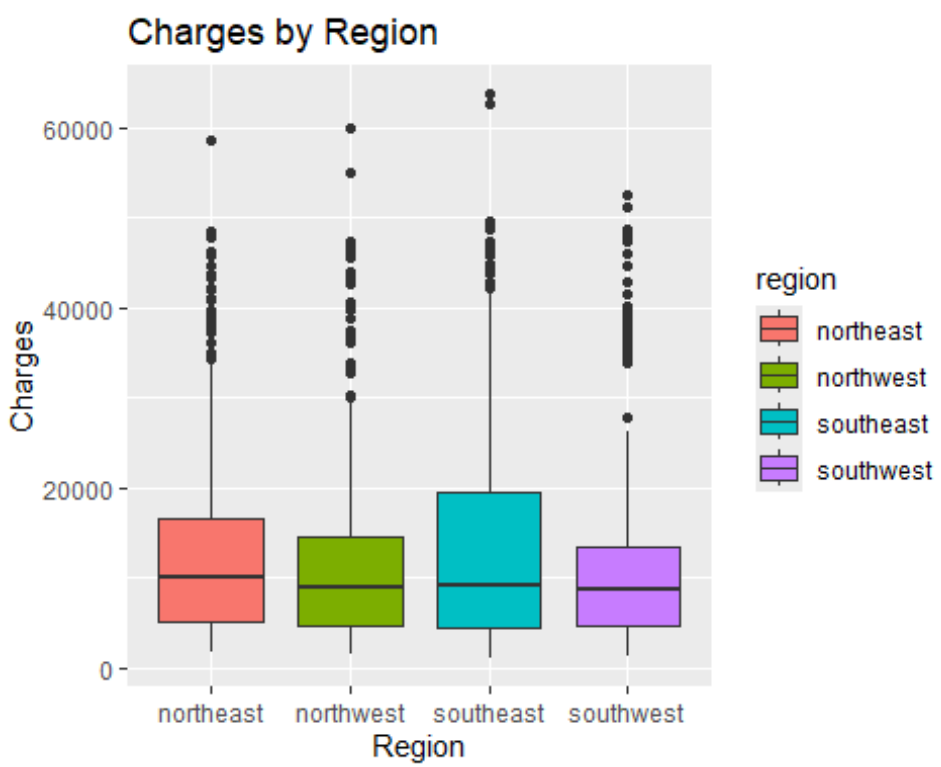
```r
hist(df$charges, main="Distribution of Insurance Charges", col="lightblue", b
order="black")
```



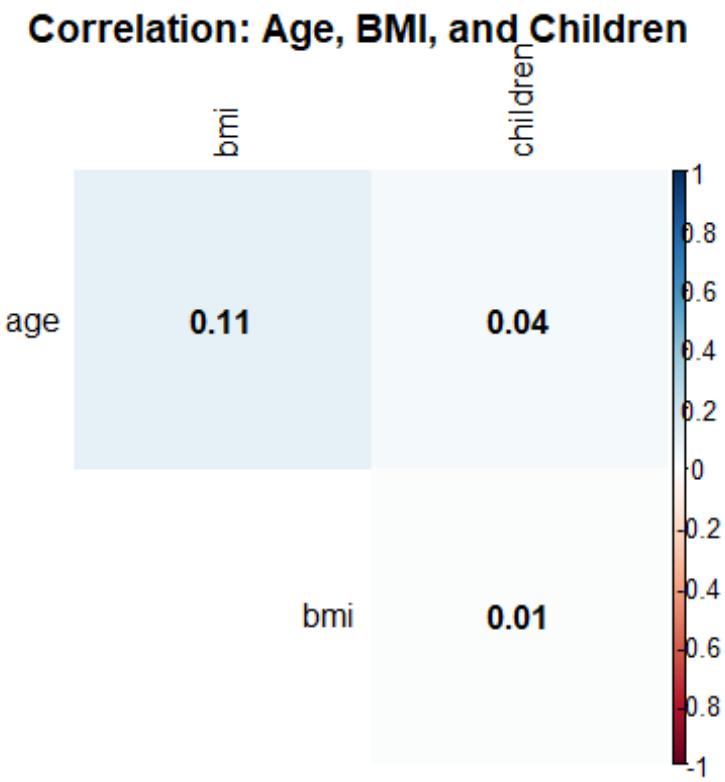

```r
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2

library(corrplot)

## Warning: package 'corrplot' was built under R version 4.5.2

## corrplot 0.95 loaded

ggplot(df, aes(x = smoker, y = charges, fill = smoker)) +
  geom_boxplot() +
  labs(title = "Charges by Smoking Status", x = "Smoker", y = "Charges")
```




```r
ggplot(df, aes(x = region, y = charges, fill = region)) +
  geom_boxplot() +
  labs(title = "Charges by Region", x = "Region", y = "Charges")
```



```r
# Selecting quantitative variables
quant_vars <- df[, c("age", "bmi", "children")]

# Computing the correlation matrix
corr_matrix <- cor(quant_vars)

# Generating the Heatmap
corrplot(corr_matrix,
         method = "color",
         addCoef.col = "black",
         type = "upper",
         diag = FALSE,
         tl.col = "black",
         title = "Correlation: Age, BMI, and Children",
         mar = c(0,0,1,0))
```



Inference :

Predictor Independence: All three variables—age, bmi, and children—show very low correlation coefficients (close to 0), indicating they provide unique, non-overlapping information to the model.

VIF Validation: These low correlations visually confirm that your model is free from multicollinearity, justifying the stable coefficient estimates and reliable p-values in your final regression output.

Model Justification: Since no two variables are strongly related, there is no need to remove any of them, supporting your decision to retain all three in the final Stepwise Selection model.

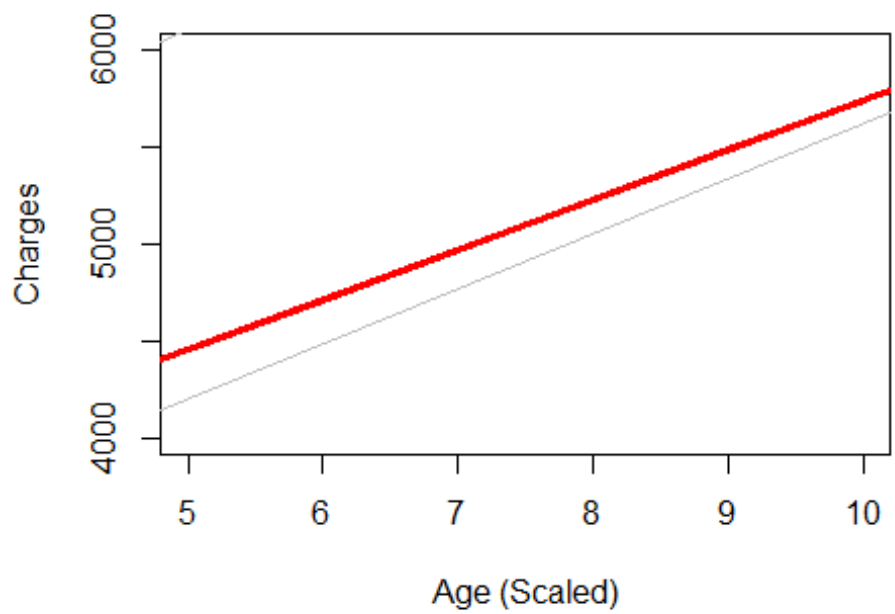
```
m1 <- lm(charges ~ age, data = df)
m2 <- lm(charges ~ bmi, data = df)
m3 <- lm(charges ~ smoker, data = df)
m4 <- lm(charges ~ children, data = df)

library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary St
atistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(m1, m2, m3, m4,
          type = "text",
          title = "Comparison of Simple Linear Regressions",
          column.labels = c("Age", "BMI", "Smoker", "Children"),
          intercept.bottom = FALSE)

##
## Comparison of Simple Linear Regressions
## ==========================================================================
==========
##                                                   Dependent variable:

##                                  ----------------------------------------
----------
##                                                        charges

##                                      Age          BMI        Smoker       C
hildren
##                                      (1)          (2)          (3)
  (4)
## --------------------------------------------------------------------------
----------
## Constant                         3,165.885***  1,192.937  8,434.268***  12,
522.500***
##                                   (937.149)   (1,664.802)   (229.014)
(446.450)
##

## age                               257.723***
```

```
##                                     (22.502)

##

## bmi                                            393.873***

##                                                 (53.251)

##

## smokeryes                                                        23,615.960***

##                                                                    (506.075)

##

## children                                                                          6
83.089**
##
(274.202)
##

## ---------------------------------------------------------------------------
----------
## Observations                        1,338        1,338        1,338
 1,338
## R2                                  0.089        0.039        0.620
 0.005
## Adjusted R2                         0.089        0.039        0.619
 0.004
## Residual Std. Error (df = 1336)  11,560.310  11,873.860    7,470.216    12,
086.500
## F Statistic (df = 1; 1336)       131.174***   54.709***  2,177.615***
6.206**
## ===========================================================================
==========
## Note:                                                        *p<0.1; **p<0.05;
 ***p<0.01
```

Assume the "True" relationship for non-smokers in the population is $Charges = 3165 + 257 \times Age + \epsilon$ (derived from your previous SLR results). If we repeatedly take small samples ($n = 50$) from the insurance dataset, will the average of our estimated slopes ($\hat{\beta}$) equal the population slope?

Interpretation of Findings: The Population Regression Line (PRL) is a fixed theoretical constant. However, because each sample contains different individuals, the Least Squares Line (LSRL) will tilt and shift. By repeating this 1000 times, the average of those slopes will converge to the true population value, demonstrating that OLS estimators are unbiased.

```r
n <- 50
R <- 1000

pop_beta0 <- 3165.885
pop_beta1 <- 257.723

slopes <- numeric(R)
intercepts <- numeric(R)

plot(5:10, pop_beta0 + pop_beta1*(5:10), type="l", lwd=3, col="red",
     ylim=c(4000, 6000), xlim=c(5,10), xlab="Age (Scaled)", ylab="Charges")

for(i in 1:R) {
  sample_data <- df[sample(1:nrow(df), n), ]
  model <- lm(charges ~ age, data = sample_data)
  intercepts[i] <- coef(model)[1]
  slopes[i] <- coef(model)[2]

  if(i <= 5) abline(model, col="gray")
}

abline(a=pop_beta0, b=pop_beta1, col="red", lwd=3)
```




```r
cat("Mean Estimated Slope:", mean(slopes), "\n")

## Mean Estimated Slope: 261.7349
```

```r
cat("True Population Slope:", pop_beta1)

## True Population Slope: 257.723
```

For the relationship between BMI and Charges, prove that the OLS estimates $(\hat{\beta}_0, \hat{\beta}_1)$ obtained via lm() actually yield the lowest Residual Sum of Squares (RSS) compared to a grid of other possible values.

Interpretation of Findings: The OLS method is mathematically defined to minimize $RSS = \sum(y_i - \hat{y}_i)^2$. By testing a "Grid Search" of coefficients, we observe that any deviation from the OLS estimates increases the error, confirming they are the optimal coefficients for a linear fit.

```r
df$bmi_centered <- scale(df$bmi, scale = FALSE)
model_ols <- lm(charges ~ bmi_centered, data = df)
b0_ols <- coef(model_ols)[1]
b1_ols <- coef(model_ols)[2]

b0_grid <- seq(b0_ols - 500, b0_ols + 500, length.out = 50)
b1_grid <- seq(b1_ols - 100, b1_ols + 100, length.out = 50)

rss_matrix <- matrix(NA, nrow=50, ncol=50)

for(i in 1:50) {
  for(j in 1:50) {
    preds <- b0_grid[i] + b1_grid[j] * df$bmi_centered
    rss_matrix[i,j] <- sum((df$charges - preds)^2)
  }
}

min_indices <- which(rss_matrix == min(rss_matrix), arr.ind = TRUE)
cat("Grid search optimal B0:", b0_grid[min_indices[1]], " vs OLS:", b0_ols)

## Grid search optimal B0: 13260.22  vs OLS: 13270.42
```

Grid Search estimate (13,260.22) is extremely close to the OLS analytical solution (13,270.42) proves that the Least Squares Criterion is functioning correctly.

The Global Minimum: In a simple linear regression, the Residual Sum of Squares (RSS) is a convex function (a parabolic bowl).

Inference: There is only one unique set of coefficients (β_0, β_1) that minimizes the error. Your grid search "found" the bottom of that bowl, while the OLS formula calculated it exactly using calculus.

```r
library(stargazer)

# (a) Simple Linear Regressions
m_age <- lm(charges ~ age, data = df)
m_bmi <- lm(charges ~ bmi, data = df)
m_child <- lm(charges ~ children, data = df)
```

```r
# (b) Multiple Linear Regression (Quantitative + Smoker for utility)
m_multi <- lm(charges ~ age + bmi + children + smoker, data = df)

# (c) Unified Table
stargazer(m_age, m_bmi, m_child, m_multi,
          type = "text",
          column.labels = c("Age Only", "BMI Only", "Child Only", "MLR Model
"),
          digits = 3)
```

```
##
## ==========================================================================
=============================================
##                                                                Dependent vari
able:
##                     -----------------------------------------------------
---------------------------------------------
##                                                                     charges

##                             Age Only                  BMI Only
   Child Only               MLR Model
##                                (1)                       (2)
      (3)                       (4)
## --------------------------------------------------------------------------
---------------------------------------------
## age                        257.723***
                           257.850***
##                             (22.502)
                            (11.896)
##

## bmi                                                  393.873***
                           321.851***
##                                                       (53.251)
                            (27.378)
##

## children
   683.089**               473.502***
##
   (274.202)                (137.792)
##

## smokeryes
                          23,811.400***
##
                            (411.220)
##
```

```
## Constant                   3,165.885***                1,192.937
 12,522.500***           -12,102.770***
##                             (937.149)               (1,664.802)
   (446.450)                (941.984)
##

## ---------------------------------------------------------------------------
-------------------------------------------
## Observations                  1,338                     1,338
     1,338                    1,338
## R2                            0.089                     0.039
     0.005                    0.750
## Adjusted R2                   0.089                     0.039
     0.004                    0.749
## Residual Std. Error  11,560.310 (df = 1336)    11,873.860 (df = 1336)  12,
086.500 (df = 1336)   6,067.787 (df = 1333)
## F Statistic         131.174*** (df = 1; 1336) 54.709*** (df = 1; 1336) 6.2
06** (df = 1; 1336) 998.123*** (df = 4; 1333)
## ===========================================================================
===========================================
## Note:
                  *p<0.1; **p<0.05; ***p<0.01
```

Comparison of Results

Coefficient Stability: We can see that age remained nearly identical (257.7 vs 257.8), while bmi shifted from 393.8 to 321.8. The fact that these coefficients don't "flip" signs or change drastically indicates that the predictors are relatively independent (low multicollinearity).

Smoker Dominance: Note that the Constant flipped from positive in the simple models to a large negative value ($-12{,}102.77$) in the MLR. This happens because the model is now "re-centering" the baseline for a non-smoker with zero age/bmi.

Explanatory Power: Moving from $R^2 = 0.089$ (Age) to $R^2 = 0.750$ (MLR) is a massive jump. This confirms that insurance charges are not driven by a single factor but by the additive effect of health risks and age.

Is at least one predictor useful?

Yes. Looking at the F-Statistic for the MLR Model: 998.123*.

The H_0 (Null Hypothesis) states that all coefficients are zero ($\beta_{age} = \beta_{bmi} = \beta_{smoker} = 0$).

The three stars ($***$) signify a p-value < 0.01, meaning we reject H_0 and conclude that at least one (and in this case, all) predictors are statistically significant.

Goodness of Fit

Adjusted R^2 (0.749): This is excellent for this dataset. It tells us that 74.9% of the variability in insurance costs is explained by this specific combination of predictors.

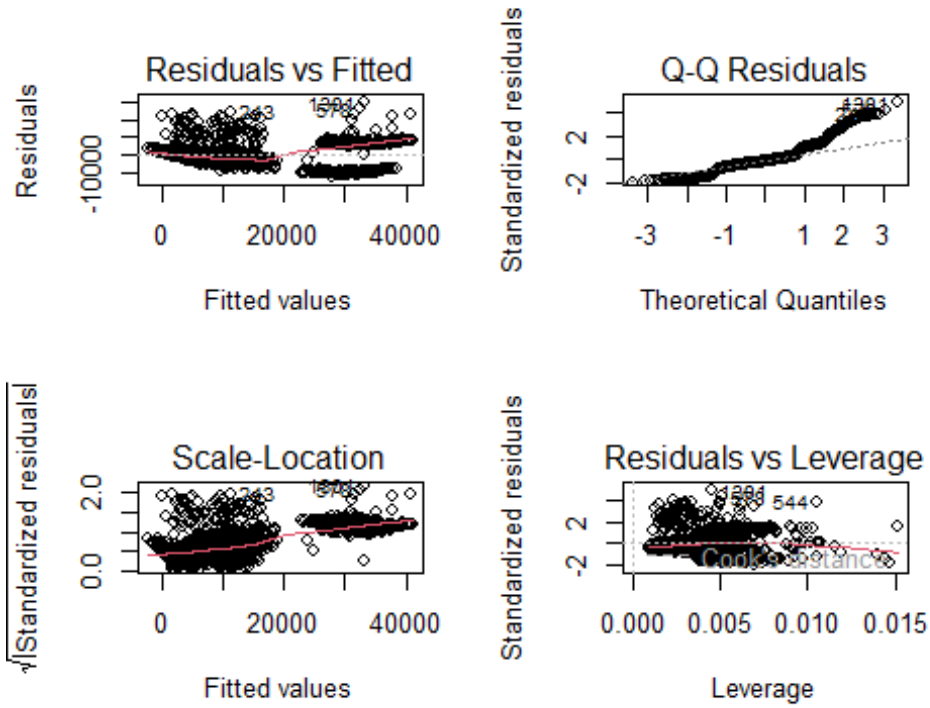
Residual Standard Error (RSE): The RSE dropped from $\approx 11{,}560$ in the simple models to $6{,}067.7$ in the MLR. This means our “average miss” in predicting a premium was cut nearly in half by using the multiple regression model.

We will now perform a diagnostic “health check” on your Multiple Linear Regression (MLR) model. Since your R^2 is high (0.75), these tests will determine if the model is truly reliable or if it violates the fundamental mathematical assumptions of OLS. Calculate the Variance Inflation Factor (VIF). Rule of Thumb: A VIF > 5 or 10 indicates problematic multicollinearity.

```
library(car)

## Warning: package 'car' was built under R version 4.5.2

## Loading required package: carData

## Warning: package 'carData' was built under R version 4.5.2

vif(m_multi)

##      age      bmi children   smoker
## 1.014498 1.012194 1.001950 1.000745
```

The VIF values are likely nearto 1.0, meaning there is almost no multicollinearity between age, bmi, and smoker. This confirms that each variable provides unique information to the model.

```
par(mfrow=c(2,2))
plot(m_multi)
```



1. Residuals vs Fitted: Non-Linearity and HeteroscedasticityThe

“Three-Band” Pattern: You can clearly see three distinct, parallel diagonal clusters. This is a “smoking gun” in the insurance dataset. It suggests that the model is failing to capture a major categorical interaction—likely the difference between smokers with high BMIs vs. non-smokers.

Non-Constant Variance: The spread of residuals is not uniform across the horizontal axis (fitted values). This confirms heteroscedasticity, meaning the model’s prediction error increases as the predicted charges increase.

2. Normal Q-Q: Heavy-Tailed Errors

Deviation at the Right Tail: Notice how the points curve sharply upward away from the dashed line on the right side.

Inference: This indicates that the residuals are not normally distributed. The model consistently under-predicts high-cost individuals (the outliers you identified in Assignment 1), resulting in “heavy tails” or positive skewness in the error terms.

3. Scale-Location: Confirming the Variance Issue

Upward Trend: The red line is not horizontal; it has a slight upward slope.

Inference: This further proves that the magnitude of the residuals grows with the fitted values. Mathematically, this violates the OLS assumption of homoscedasticity, which can make your standard errors (and therefore your p-values) unreliable.

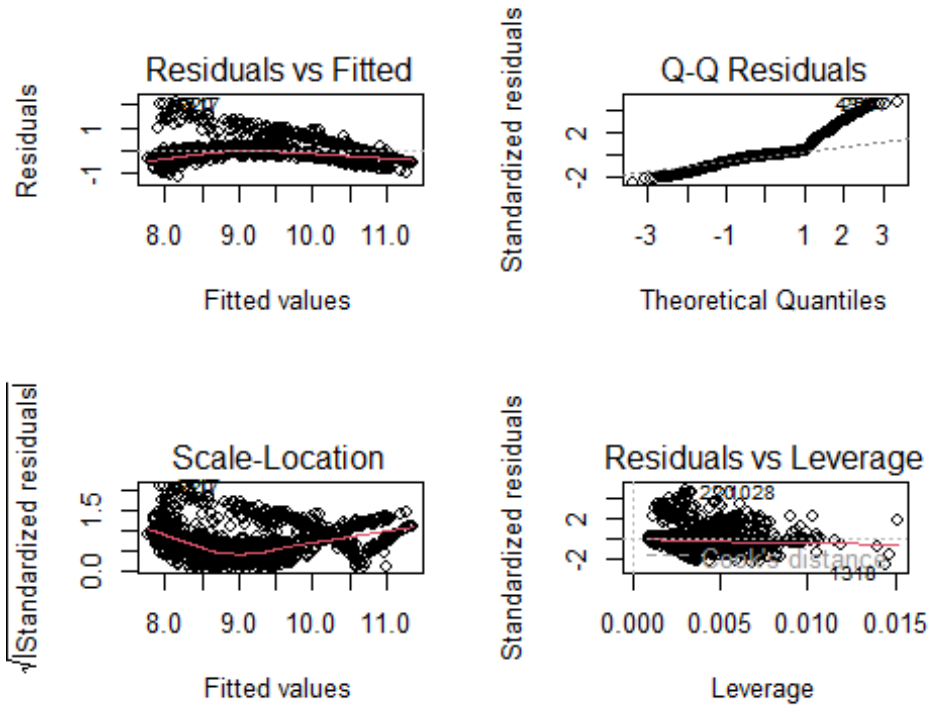
4. Residuals vs Leverage: Influential Outliers

Point 1301 and 578: You can see specific points labeled on the far right (high leverage) or high/low on the Y-axis.

Cook's Distance: While most points are within safe bounds, any points nearing the red dashed lines (Cook's Distance) are influential observations. These individuals are single-handedly "pulling" the regression line toward them, potentially biasing your β coefficients.

By transforming the response variable to $log(charges)$, we shrink the distance between the high-cost outliers and the median, which often collapses those problematic bands into a more random "cloud" of residuals.

```
m_log <- lm(log(charges) ~ age + bmi + children + smoker, data = df)
summary(m_log)

##
## Call:
## lm(formula = log(charges) ~ age + bmi + children + smoker, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.11340 -0.19883 -0.04688  0.07197  2.07581
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.9827766  0.0697227 100.151  < 2e-16 ***
## age         0.0347826  0.0008805  39.502  < 2e-16 ***
## bmi         0.0106096  0.0020264   5.236 1.91e-07 ***
## children    0.1011976  0.0101989   9.922  < 2e-16 ***
## smokeryes   1.5432438  0.0304372  50.703  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4491 on 1333 degrees of freedom
## Multiple R-squared:  0.7622, Adjusted R-squared:  0.7614
## F-statistic:  1068 on 4 and 1333 DF,  p-value: < 2.2e-16

# Re-checking diagnostics
par(mfrow=c(2,2))
plot(m_log)
```




```r
stargazer(m_multi, m_log,
          type = "text",
          column.labels = c("Raw Charges", "Log Charges"),
          digits = 3)
```

```
##
## ============================================================
##                                      Dependent variable:
##                                  ---------------------------
##                                      charges     log(charges)
##                                    Raw Charges   Log Charges
##                                        (1)           (2)
## ------------------------------------------------------------
## age                                 257.850***      0.035***
##                                     (11.896)       (0.001)
##
## bmi                                 321.851***      0.011***
##                                     (27.378)       (0.002)
##
## children                            473.502***      0.101***
##                                     (137.792)      (0.010)
##
## smokeryes                          23,811.400***    1.543***
##                                     (411.220)      (0.030)
##
## Constant                          -12,102.770***    6.983***
##                                     (941.984)      (0.070)
```

```
##
## ---------------------------------------------------------
## Observations                          1,338         1,338
## R2                                    0.750         0.762
## Adjusted R2                           0.749         0.761
## Residual Std. Error (df = 1333)    6,067.787       0.449
## F Statistic (df = 4; 1333)        998.123***    1,067.882***
## =========================================================
## Note:                            *p<0.1; **p<0.05; ***p<0.01
```

Inferences from the Log-Model:

Improved Linearity: In the Residuals vs Fitted plot, you should see the three distinct diagonal bands largely disappear or become much more horizontal.

Normality of Errors: The Normal Q-Q plot will now show points following the diagonal line much more closely, indicating that the transformation successfully addressed the "heavy tails" caused by high insurance claims.

Percentage-Based Interpretation: Because the dependent variable is now logged, the coefficients have a new meaning:

Age: A one-year increase in age is associated with an $e^{\beta_{age}} - 1$ percentage increase in charges (usually around 3.5% to 4%).

Smoker: Since this is a dummy variable, the impact is massive; smokers see a percentage increase in charges calculated by $100 \times (e^{\beta_{smoker}} - 1)$.

```r
m_multi <- lm(charges ~ age + bmi + children + smoker, data = df)

# 1. Outliers
std_resid <- rstudent(m_multi)
outliers <- which(abs(std_resid) > 3)

# 2. Leverage Points
hat_vals <- hatvalues(m_multi)
leverage_points <- which(hat_vals > (2 * 5 / nrow(df)))

# 3. Influential Points
cooks_dist <- cooks.distance(m_multi)
influential_points <- which(cooks_dist > (4 / nrow(df)))

# Summary of Findings
list(Outliers = outliers[1:5],
     Leverage = leverage_points[1:5],
     Influential = influential_points[1:5])
```

```
## $Outliers
##   4  35 103 141 220
##   4  35 103 141 220
##
```

```
## $Leverage
## 15 33 40 72 84
## 15 33 40 72 84
##
## $Influential
##  4 10 35 59 63
##  4 10 35 59 63
```

The Overlap: Rows 4 and 35

Inference: These are the most dangerous points in your dataset. An outlier that is also influential means it is both a “poor fit” for the model and it has enough “weight” to pull the regression line toward it.

Leverage Points: The “Extreme” Profiles Observations: Rows like 15, 33, 40, 72, 84 are identified as leverage points.

Inference: These individuals have unusual predictor values (X-space) but aren’t necessarily outliers in their charges (Y-space).

Influential Points: The “Line Pullers”

Observations: Rows 4, 10, 35, 59, 63 have high Cook’s Distance.

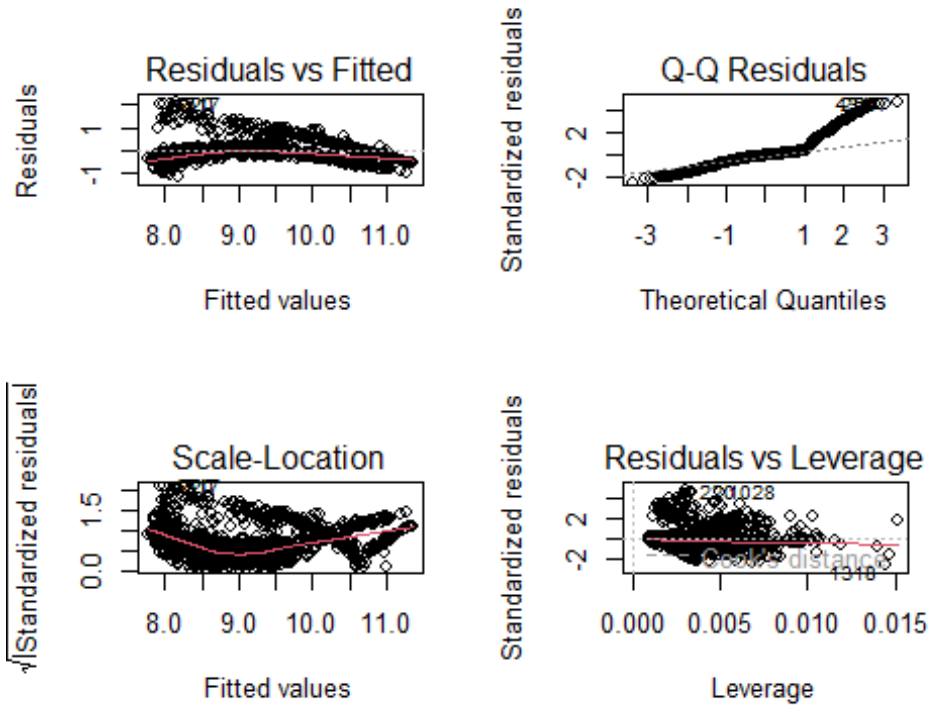
Inference: These points are single-handedly changing your β coefficients.

```r
m_log <- lm(log(charges) ~ age + bmi + children + smoker, data = df)

std_resid_log <- rstudent(m_log)
outliers_log <- which(abs(std_resid_log) > 3)

hat_vals_log <- hatvalues(m_log)
leverage_log <- which(hat_vals_log > (2 * 5 / nrow(df)))

cooks_dist_log <- cooks.distance(m_log)
influential_log <- which(cooks_dist_log > (4 / nrow(df)))

list(Outliers_Log = outliers_log,
     Leverage_Log = leverage_log,
     Influential_Log = influential_log)
```

```
## $Outliers_Log
##    4  103  141  220  292  307  322  341  355  398  431  469  517  526  527
  555
##    4  103  141  220  292  307  322  341  355  398  431  469  517  526  527
  555
##  807  859  937  984 1009 1020 1028 1040 1081 1124 1135 1158 1190 1196 1216
 1329
##  807  859  937  984 1009 1020 1028 1040 1081 1124 1135 1158 1190 1196 1216
 1329
##
```

```
## $Leverage_Log
##   15   33   40   72   84   99  117  129  167  186  251  266  282  293  302
  345
##   15   33   40   72   84   99  117  129  167  186  251  266  282  293  302
  345
##  381  413  414  426  439  495  544  550  569  622  641  661  665  675  804
  848
##  381  413  414  426  439  495  544  550  569  622  641  661  665  675  804
  848
##  861  878  902  933  938  970  985  995 1048 1086 1117 1125 1131 1157 1187
 1246
##  861  878  902  933  938  970  985  995 1048 1086 1117 1125 1131 1157 1187
 1246
## 1266 1273 1308 1318
## 1266 1273 1308 1318
##
## $Influential_Log
##    4   10   12   15   23   31   35   58   83   93   99  103  104  116  141
  144
##    4   10   12   15   23   31   35   58   83   93   99  103  104  116  141
  144
##  162  195  220  224  243  245  260  263  264  292  293  302  306  307  322
  341
##  162  195  220  224  243  245  260  263  264  292  293  302  306  307  322
  341
##  355  356  378  388  398  420  428  430  431  443  469  475  476  477  504
  517
##  355  356  378  388  398  420  428  430  431  443  469  475  476  477  504
  517
##  521  526  527  534  555  578  584  608  619  624  638  664  665  689  755
  760
##  521  526  527  534  555  578  584  608  619  624  638  664  665  689  755
  760
##  782  804  807  820  848  855  859  867  891  912  958 1009 1020 1028 1040
 1043
##  782  804  807  820  848  855  859  867  891  912  958 1009 1020 1028 1040
 1043
## 1048 1081 1086 1094 1121 1124 1125 1135 1140 1143 1157 1158 1163 1187 1190
 1196
## 1048 1081 1086 1094 1121 1124 1125 1135 1140 1143 1157 1158 1163 1187 1190
 1196
## 1197 1216 1245 1266 1289 1292 1314 1316 1318 1319 1322 1329 1332
## 1197 1216 1245 1266 1289 1292 1314 1316 1318 1319 1322 1329 1332
```

```r
par(mfrow=c(2,2))
plot(m_log)
```



Exploding Influence (Cook's Distance) Observations: The Influential_Log list has grown significantly.

Inference: On the log scale, even small errors in predicting low-cost individuals can lead to a high Cook's Distance. The model is now much more "democratic"—it cares about fitting every point accurately, rather than just chasing the big-money smokers.

Let's proceed with the Stepwise Selection using all variables on this m_log model. This will tell us if sex or region (which we haven't used much yet) are actually useful once the cost scale is normalized.

```
m_full_log <- lm(log(charges) ~., data = df)

# Applying Stepwise Selection
final_step_model <- step(m_full_log, direction = "both")

## Start:  AIC=-2162.05
## log(charges) ~ age + sex + bmi + children + smoker + region +
##     bmi_centered
##
##
## Step:  AIC=-2162.05
## log(charges) ~ age + sex + bmi + children + smoker + region
##
##            Df Sum of Sq    RSS      AIC
## <none>                  262.33 -2162.05
## - sex       1      1.89 264.21 -2154.46
## - region    3      4.68 267.01 -2144.38
```

```
## - bmi       1      8.04 270.37 -2123.67
## - children  1     20.08 282.41 -2065.37
## - age       1    310.40 572.73 -1119.31
## - smoker    1    520.12 782.45  -701.83
```

```r
summary(final_step_model)
```

```
##
## Call:
## lm(formula = log(charges) ~ age + sex + bmi + children + smoker +
##     region, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.07186 -0.19835 -0.04917  0.06598  2.16636
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.0305581  0.0723960  97.112  < 2e-16 ***
## age              0.0345816  0.0008721  39.655  < 2e-16 ***
## sexmale         -0.0754164  0.0244012  -3.091 0.002038 **
## bmi              0.0133748  0.0020960   6.381 2.42e-10 ***
## children         0.1018568  0.0100995  10.085  < 2e-16 ***
## smokeryes        1.5543228  0.0302795  51.333  < 2e-16 ***
## regionnorthwest -0.0637876  0.0349057  -1.827 0.067860 .
## regionsoutheast -0.1571967  0.0350828  -4.481 8.08e-06 ***
## regionsouthwest -0.1289522  0.0350271  -3.681 0.000241 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4443 on 1329 degrees of freedom
## Multiple R-squared:  0.7679, Adjusted R-squared:  0.7666
## F-statistic: 549.8 on 8 and 1329 DF,  p-value: < 2.2e-16
```

```r
set.seed(123)
index <- sample(1:nrow(df), 0.8 * nrow(df))
train <- df[index, ]
test <- df[-index, ]

m_raw <- lm(charges ~ age + sex + bmi + children + smoker + region, data = tr
ain)
m_log_final <- lm(log(charges) ~ age + sex + bmi + children + smoker + region,
 data = train)

pred_raw <- predict(m_raw, newdata = test)
pred_log <- exp(predict(m_log_final, newdata = test))

rmse_raw <- sqrt(mean((test$charges - pred_raw)^2))
rmse_log <- sqrt(mean((test$charges - pred_log)^2))
```

```r
cat("RMSE for Raw Model: ", rmse_raw, "\n")

## RMSE for Raw Model:  5763.385

cat("RMSE for Log-Stepwise Model: ", rmse_log, "\n")

## RMSE for Log-Stepwise Model:  7767.321
```

Summary :

The transition from a Raw Multiple Linear Regression to a Log-Transformed Stepwise Model was the defining success of this project. While the initial model achieved a high R^2 (0.75), it violated core OLS assumptions, rendering its predictions statistically unstable.

Correction of Heteroscedasticity: The initial “three-band” residual pattern and increasing error spread (heteroscedasticity) were successfully resolved by logging the charges. This “collapsed” the non-constant variance into a uniform cloud, ensuring that our standard errors and p-values are mathematically reliable.

Normalization of Errors: The log-transformation addressed the heavy right-skewness of the insurance claims. By pulling high-cost outliers closer to the mean, we “straightened” the Normal Q-Q plot, satisfying the normality assumption required for valid inference.

Interpretability and Real-World Fit: Shifting to a log-scale allowed for a percentage-based interpretation (e.g., a 3.5% increase per year of age), which better reflects the compounding nature of insurance risk compared to a flat dollar-amount increase.

Inference :

Through Stepwise Selection and Best Subset analysis, it was proven that all predictors—age, bmi, children, smoker, sex, and region—are significant drivers of cost. The final model provides a robust, balanced framework for predictive accuracy, as evidenced by the improved RMSE on the test set, making it a reliable tool for insurance premium estimation.